



GRADUATION INTERNSHIP REPORT

**Topic: “DEVELOPING A RAG SYSTEM AND AI
CHATBOT TO SUPPORT EMPLOYEES ACROSS
DEPARTMENTS”**

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Class : E21CQCNTT01-N
Course : 2021-2026
Major : Information Technology
Program : High-quality program

Ho Chi Minh City, August 2025



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Acknowledgement!

I have chosen the topic "Developing an Application to Support Employees Across Departments using an AI Chatbot and RAG Technique" to complete my Graduation Internship at the Posts and Telecommunications Institute of Technology, Ho Chi Minh City Campus.

To complete this project, I would like to express my sincere gratitude to my advisor, M.S. Nguyen Trung Hieu from the Faculty of Information Technology 2, for his dedicated support, guidance, and assistance throughout the process of completing this program and project.

This project focuses on building an Artificial Intelligence application, researching Large Language Models and the RAG technique. With a user-friendly interface and useful features, the application aims to help users save research time, improve efficiency, and enhance the quality of work that requires complex reasoning and confidentiality.

Throughout the project, our team has learned and applied the knowledge acquired from subjects such as Web Programming, System Analysis and Design, Database Management, Artificial Intelligence, Natural Language Processing, Machine Learning, and Deep Learning to complete the project. This process has not only helped us improve our technical skills but also enhanced our teamwork, problem-solving, and effective time management abilities. More importantly, it has allowed us to apply an Artificial Intelligence program to practical work in the various departments of our internship company.

Once again, our team would like to extend our deepest thanks.

Ho Chi Minh City, August 18, 2025

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GLOSSARY OF TERMS

Table 1: Glossary Of Terms

Thuật ngữ	Viết đầy đủ	Giải thích
AI	Artificial Intelligence	A field of computer science that studies and develops systems capable of performing tasks that require human intelligence.
AGI	Artificial General Intelligence	An AI system that can perform any intellectual task a human can.
ANI	Artificial Narrow Intelligence	AI designed to perform a specific task.
API	Application Programming Interface	A set of definitions and protocols that allows applications to communicate with each other.
ASI	Artificial Superintelligence	Intelligence surpasses human intelligence in every field.
ChromaDB	Chroma Database	A vector database is used to store and search embedded data.
DL	Deep Learning	A machine learning method that uses multi-layered neural networks to analyze complex data.
FAQ	Frequently Asked Questions	A list of common questions.
JSON	JavaScript Object Notation	A lightweight, easy-to-read/write data format used for data exchange between systems.
LLM / LLMs	Large Language Model(s)	An AI model trained in a massive amount of text data, capable of processing and generating natural language.
ML	Machine Learning	A subfield of AI that allows computers to learn from data without explicit programming.
NLP	Natural Language Processing	A field of AI focused on the interaction between computers and human language.
RAG	Retrieval-Augmented Generation	A technique that enhances answer generation by retrieving information from external data sources.
RLHF	Reinforcement Learning from Human Feedback	A method for fine-tuning AI models based on human evaluation.
UI	User Interface	The visual part that allows users to interact with a system.
Vector Database	Vector Database	A database that stores vector embeddings for semantic search.

INTRODUCTION

1) Project Overview: "Developing An Application To Support Employees Across Departments Using An Ai Chatbot And Rag Technique"

In the context of the 4.0 Industrial Revolution and the robust trend of digital transformation, particularly with the rapid advancements in computer science within the fields of artificial intelligence and Large Language Models, modern enterprises are accumulating a massive volume of internal data and knowledge. These documents exist in various formats, including analytical reports, business processes, company policies, project documents, contracts, internal guides, and emails. However, the ability to quickly and accurately leverage and retrieve information from this vast knowledge base has become a significant challenge, leading to time wastage and negatively impacting employee productivity.

The project, "Developing an Application to Support Employees Across Departments using an AI Chatbot and RAG Technique," was undertaken to directly address this problem. By combining the power of Large Language Models (LLMs) with the advanced Retrieval-Augmented Generation (RAG) technique, the project aims to build an intelligent chatbot system that serves as a professional virtual assistant for employees.

The core of this system is its ability to "comprehend" and retrieve information in real time directly from the enterprise's internal document repository. Instead of providing answers based on statically trained knowledge, the chatbot can deliver accurate, up-to-date, and well-sourced responses from the provided documents. This system is not merely a passive information retrieval tool but rather a digital colleague that helps employees quickly find necessary data, understand complex procedures, and make decisions based on reliable information, thereby enhancing the overall productivity and efficiency of the entire organization.

2) Justification For The Project

The selection of the project "Developing an Application to Support Employees Across Departments using an AI Chatbot and RAG Technique" stems from a combination of practical business needs, the immense potential of emerging technologies, and the team's professional development goals. Specifically, the key reasons include:

- **The Urgency of Internal Knowledge Management:** Through our internship and observation of the modern workplace, our team identified a common challenge in many enterprises, especially traditional ones: employees, particularly new hires, spend a significant amount of time and effort searching for specific information within thousands of internal documents. This "information overload" coupled with the difficulty of finding accurate data not only reduces individual productivity but can also lead to avoidable errors. Developing an intelligent support tool to solve this problem is a real and urgent need.

- **The Breakthrough Potential of RAG Technology:** The world is witnessing a strong wave of development in Generative AI and Large Language Models (LLMs). However, applying LLMs in an enterprise environment faces two major barriers: the phenomenon of hallucinations (fabricating information) and proprietary data security issues. The RAG (Retrieval-Augmented Generation) technique has emerged as a groundbreaking solution to these problems. It allows LLMs to access a reliable and controlled source of knowledge, ensuring that answers are both intelligent, accurate, and secure. Choosing RAG demonstrates a keen awareness and alignment with the most advanced technological trends.

Practical Applicability and High Scalability: The solution developed in this project has high practical applicability and can be deployed at any enterprise seeking to optimize workflows and knowledge management. The system can start by supporting a few strategic departments and can be easily scaled to the entire organization. Creating a product with real practical value that can be used immediately upon completion is one of our team's key objectives.

Academic Value and In-depth Skill Development: This project offers a valuable opportunity for the team to research and master one of the most complex and effective AI architectures available today. Its execution requires the application of comprehensive knowledge from multiple fields: Natural Language Processing (NLP), AI system architecture, working with Vector Databases, and complete application development skills. These are highly valuable and in-demand skills in the current job market.

Based on these reasons, our team believes this project not only meets the requirements for a graduation internship project but also delivers practical value, anticipates technological trends, and opens up many avenues for future development.

3) Current Situation

To address the need for internal knowledge retrieval and management, enterprises currently employ various solutions with differing levels of effectiveness. However, most of these solutions have inherent limitations.

a. Manual Storage and Retrieval Systems:

- **Description:** This is the most common method, involving the use of shared drives (e.g., Google Drive, OneDrive), internal file servers, or cloud storage platforms like Microsoft SharePoint, as observed at our internship company. Employees manually search for documents by navigating through folder trees or using basic search functions.
- **Limitations:**

Keyword-based, not semantic, search: The search function only returns documents containing the exact keywords, lacking the ability to understand context or synonyms. For instance, a search for "Q4 revenue report" might miss a document named "Year-end business summary."

Information overload: A simple query can return dozens or even hundreds of documents, forcing users to manually read and filter to find the information they need, which is a significant waste of time.

Version control issues: Managing document versions (v1, v2, final, final_edited...) is a nightmare, and users can easily end up using outdated information.

Human dependency: The system's effectiveness is entirely dependent on employees' diligence and discipline in organizing documents.

b. Traditional Rule-based Chatbots:

- **Description:** Some companies deploy basic chatbots to answer frequently asked questions (FAQs). These chatbots operate based on a pre-programmed set of rules and scripts (if-then logic).

- **Limitations:**

Lack of flexibility: The chatbot can only answer predefined questions. Any variation in the structure or vocabulary of a query will cause it to fail.

Poor scalability: Adding a new question-answer pair requires manual intervention from the technical team, which is time-consuming and costly to maintain.

Inability to process documents: These chatbots are incapable of "reading" and extracting information from text files, PDFs, or reports.

c. Enterprise Search Platforms:

- **Description:** These are more advanced solutions (e.g., Elasticsearch, Microsoft Search) capable of indexing a large volume of data from various sources.

- **Limitations:**

Retrieval tool, not an answer tool: Although more powerful, their core function remains returning a list of the most relevant documents. The cognitive load is still placed on the user, who must open and read the documents to find the final answer.

Lack of synthesis: They cannot synthesize information from multiple different documents to provide a single, concise answer.

Conclusion on the Current Situation: A significant "gap" is evident between existing tools and the actual needs of users. Employees do not just want to find documents; they want to get answers. Current solutions are either too manual, too rigid, or only function at the retrieval level without achieving the capability of understanding and synthesizing information. This is precisely the gap that this project aims to bridge by applying an AI Chatbot and the RAG technique, offering a far more natural and effective mode of interaction.

4) Project Objectives

Based on the analysis of the current situation and the opportunities presented by technology, this project sets forth the following specific and clear objectives:

a. General Objective:

To successfully develop an intelligent **AI Chatbot** application that utilizes the Retrieval-Augmented Generation (RAG) technique to optimize the retrieval and management of knowledge from an enterprise's internal documents, enabling employees to access information quickly, accurately, and naturally.

b. Specific Objectives:

Develop a complete RAG system: Fully implement all components of a RAG pipeline, including:

Indexing: Process, chunk, and embed content from various input documents (PDF, DOCX, TXT, etc.).

Storage: Store the generated vectors in a Vector Database for efficient searching.

Retrieval & Generation: Retrieve relevant information and use a Large Language Model (LLM) to synthesize and generate the final answer.

Develop a user-friendly Chatbot interface: Build a basic web application interface with user and departmental access control, allowing employees to easily upload their own documents and departmental documents to interact with the chatbot.

Ensure accuracy and transparency: The answers generated by the chatbot must be strictly based on the provided documents to minimize hallucinations. The system must be able to cite the source documents used to generate the answer, enabling users to verify the information.

Effectively support the Vietnamese language: The entire system must be capable of proficiently handling documents and queries in Vietnamese, including issues related to punctuation, character encoding (Unicode), and grammatical nuances.

Evaluate and demonstrate effectiveness: Conduct experiments with a sample dataset (e.g., process and policy documents from a hypothetical department) to evaluate the system's performance and demonstrate the solution's effectiveness compared to traditional search methods.

5) Problem Analysis

To develop the proposed system, we must analyze the problem into its core components, including input data, key processing requirements, and desired output results.

a) Input Data:

The system needs to process two main types of input data:

- Knowledge Base: This is a collection of an enterprise's internal documents. These documents are unstructured and can be in various formats, most commonly:
 - Text Files: `.pdf`, `.docx`, `.txt`.
 - Tabular data: `.csv`, `.xlsx` (for advance scope).
- User Query: These are questions entered by a user in natural language (Vietnamese). The questions can be simple (asking for a specific piece of information) or complex (requiring synthesis or comparison).

b) Processing Requirements:

The system's processing is divided into two main flows, corresponding to the RAG architecture:

- Flow 1: Data Processing and Indexing Pipeline (one-time execution for new documents):

Load & Extract: The system must be able to read files from various formats (`.pdf`, `.docx`, etc.) and extract their raw text content.

Chunking: The raw text is then divided into smaller, meaningful segments or "chunks." This is essential because language models have a limited input context window, and chunking improves the accuracy of relevant information retrieval.

Embedding: Each text chunk is passed through an Embedding Model to be converted into a numerical vector. This vector is a sequence of numbers representing the semantic meaning of the chunk.

Store: The pairs of (vector, original text chunk) are stored and indexed in a Vector Database.

- Flow 2: Query and Generation Pipeline (executed each time a user asks a question):

Query Embedding: The user's question is also converted into a vector using the same embedding model from Flow 1.

Retrieve: The system uses the query's vector to perform a similarity search in the vector database to find the N most relevant text chunks.

Augment Prompt: The most relevant chunks are retrieved and combined with the user's original question to create a comprehensive prompt.

Generate Answer: This complete prompt is sent to a Large Language Model (LLM). The LLM uses the provided context (the chunks) to generate a coherent, accurate, and natural-sounding answer for the user.

c) Output Results:

The final output received by the user must meet the following criteria:

- **Direct Answer:** A coherent text segment that directly answers the user's question, rather than just returning a list of documents.
- **Source Citation:** The answer is accompanied by source information (e.g., document name, page number) that the system used to generate the response.

6) Team's Contribution Compared To Existing Solutions.

Although built on foundational concepts of AI and language processing, this project offers superior contributions and improvements over current knowledge management programs and solutions in enterprises:

Shifting from "Document Search" to "Answer Retrieval": This is the most significant and crucial contribution. Instead of just returning a list of relevant documents and forcing the user to read them (as with Google Drive, Microsoft SharePoint, or Enterprise Search), our system provides a direct, concise, and synthesized answer from multiple sources. This drastically saves employees time and effort.

Enhancing Semantic Understanding over Keyword Matching: Older systems operate based on keyword matching. Our solution, through the embedding model, can "understand" the meaning and intent behind the user's question. As a result, the system can find accurate information even when the user uses completely different wording from the original document.

Creating a "Living" and Flexible Knowledge System: Unlike rigid, hard-to-update rule-based chatbots, the RAG system allows for incredibly flexible knowledge updates. When a new policy, procedure, or report is available, an administrator simply adds the document to the knowledge base. The system automatically learns and can answer questions related to the new document immediately without requiring any rule-based reprogramming.

Increasing Transparency and Reliability through Source Citation: To address the hallucination problem of large AI models, our system is designed to always cite the source documents it used. This feature allows users to verify the information, building trust and confidence in the AI-provided results—a critically important factor in an enterprise environment.

Building a Complete and Specialized Solution for Vietnamese: This project goes beyond theoretical research by building a complete application, from the processing backend to the communication frontend. It is specifically optimized for handling the Vietnamese language, contributing to the application of advanced AI technology to solve practical problems in Vietnam.

TEAM MEMBER TASK ASSIGNMENTS

Table 2: Team member task assignments

No.	Main Task	Detailed Content	Responsible Member
1	Literature Review	- Study the fundamentals of AI Chatbots (Rule-based, Retrieval-based, Generative). - Research Advanced RAG Chatbot techniques (Router, Reranking, Access Control). - Review existing enterprise chatbot systems for reference.	Whole Team
2	Data Collection & Preprocessing	- Collect IT Helpdesk documents and departmental datasets. - Perform data cleaning, normalization, and preprocessing (remove special characters, ensure Unicode consistency, chunking).	Nguyen Tan Phuc
3	System Analysis & Design	- Plan the implementation of the Advanced RAG system. - Draw system design diagrams (architecture, sequence diagram, use case diagram). - Define role-based access logic and Router/Reranker integration. - Propose UI/UX design for the web application.	Whole Team
4	Building the Advanced RAG Model	- Implement the RAG pipeline: document loading, chunking, embedding, and vector storage (ChromaDB). - Connect with LLM via Groq API to generate answers. - Integrate Router & Reranker for improved accuracy.	Pham Thanh Linh
5	Backend API	- Build API endpoints using FastAPI (e.g., /ask). - Handle requests from the frontend, call the RAG pipeline, and return responses with citations.	Phạm Thanh Linh
6	Evaluation for RAG Advance	- Design test cases to measure accuracy, latency, and reliability. - Compare performance with traditional search methods. - Record and document evaluation results.	Nguyễn Tấn Phúc
7	Build App UI	- Build the web application (login, file upload, chat interface). - Integrate the frontend with backend API. - Optimize UX/UI for better user experience.	Nguyễn Tấn Phúc
8	Testing & System Evaluation	- Perform system-wide testing (frontend + backend). - Assess stability, usability, and accuracy. - Collect feedback, fix bugs, and optimize performance.	Pham Thanh Linh
9	Report Writing	- Write the final internship report (theory, system design, implementation, evaluation, future development). - Prepare slides and demo video. - Assign presentation roles for the defense session.	Whole Team

CHAPTER 1: THEORETICAL FOUNDATION

1.1 Concept of AI

Artificial Intelligence (AI) is a broad and fundamental field of computer science that focuses on studying and building computer systems capable of performing tasks that typically require human intelligence. According to John McCarthy, one of the co-founders of the term in 1956, AI is "the science and engineering of making intelligent machines."

The capabilities that AI aims to emulate include learning (acquiring information and rules for using that information), reasoning (using rules to reach approximate or definite conclusions), self-correction, perception (using artificial senses like cameras and microphones), and natural language processing.

Basically, AI can be classified based on its capabilities into three main types:

- **Artificial Narrow Intelligence (ANI):** This is the only type of AI that humans have achieved to date. Narrow AI is designed and trained to perform a specific task or a closely related set of tasks. It can outperform humans in its specialized domain but is unable to apply its knowledge outside of its programmed scope. Examples include facial recognition software, virtual assistants (Siri, Alexa), product recommendation systems on e-commerce sites, and the chatbot system developed in this project.
- **Artificial General Intelligence - AGI:** This is the next level of AI, where a machine possesses intelligence equivalent to that of a human. An AGI system can understand, learn, and apply its knowledge to solve any problem that a human can. AGI remains a research goal and has not yet been realized.
- **Artificial Superintelligence - ASI:** This is a hypothetical form of AI that surpasses the intelligence of the most brilliant human minds in every field, from scientific creativity to social intelligence.

Within the scope of this project, we focus on applying **Artificial Narrow Intelligence**, specifically techniques in the fields of **Machine Learning** and **Deep Learning**, to solve a practical problem within an enterprise.

1.2 Concept of Machine Learning

For a computer system to "understand" and "communicate" in natural language within a specialized context like employee support, it cannot rely on fixed, hard-coded rules. Instead, the system must have the ability to learn from the text data itself. **Machine Learning (ML)** is the subfield of AI that provides the methods and tools to perform this learning process.

At its core, Machine Learning enables computers to automatically recognize patterns and rules from data to make decisions or predictions, without explicit human programming. In the field of **Natural Language Processing (NLP)**, this is the key to solving complex problems.

The main Machine Learning methods, as applied in the context of this project's text processing, can be understood as follows:

- **Supervised Learning:** This is a foundational method for many language tasks. A model learns from a dataset of text that has been manually labeled by humans.
 - **Relevant Application to the Project:** Classifying documents by topic (e.g., contract, report, or procedure), machine translation, or recognizing important entities within a sentence (project names, dates, terms).
- **Unsupervised Learning:** This method is particularly useful when working with an enterprise's massive, unlabeled document repository. The model autonomously discovers the underlying structure of the data.
 - **Relevant Application to the Project:** Automatically clustering documents with similar topics (topic modeling), or more importantly, learning **word embeddings**—the foundation for a computer to understand the semantic meaning of words and sentences.
- **Reinforcement Learning:** Although less common in traditional NLP tasks, this method is becoming central to fine-tuning modern chatbots. The model learns through a process of trial and error, receiving a "reward" for good answers and a "penalty" for poor ones.
 - **Relevant Application to the Project:** Advanced techniques like **Reinforcement Learning from Human Feedback (RLHF)** are used to make chatbots (like ChatGPT) more helpful, truthful, and safe.

The development of these Machine Learning methods, especially supervised learning on massive text datasets, laid the groundwork for the emergence of **Deep Learning** and subsequently **Large Language Models (LLMs)**—the core technology applied in this project to create a truly intelligent chatbot.

1.3 Concept of Deep Learning

Deep Learning (DL) is a specialized subfield of Machine Learning that uses complex *Artificial Neural Networks* with multiple layers to analyze data. The term "deep" refers to the large number of these layers. While traditional Machine Learning models can be limited in their ability to handle the complexity of raw data (such as text or images), Deep Learning excels by automatically learning various levels of abstract representations.

For the natural language processing task in this project, the power of Deep Learning is demonstrated by:

- The model's ability to automatically learn from individual characters and words in the initial layers.
- In deeper layers, it combines this knowledge to recognize phrases and grammatical structures.
- In the deepest layers, it can grasp the semantic meaning, nuance, and complex relationships within an entire passage of text.

One of the most significant and revolutionary advantages of Deep Learning over traditional Machine Learning is automatic feature extraction. Instead of engineers painstakingly designing features for text data (like word frequency counts, n-grams), Deep Learning models can learn the most important features directly from the input data itself, optimizing performance and minimizing human intervention.

The true breakthrough in applying Deep Learning to NLP came from an architecture called the Transformer, introduced in 2017. The core of the Transformer is the attention mechanism, which allows the model, when processing a word, to "look back" and weigh the importance of all other words in the sentence. This helps it effectively capture context and long-range dependencies, a weakness of previous architectures.

The Transformer architecture is the foundation for almost all modern Large Language Models (LLMs). It is thanks to the power of Deep Learning and Transformer architecture that researchers have been able to build colossal language models, creating a revolution in how computers interact with humans. We will explore LLMs in more detail in the next section.

1.4 The Evolution of LLMs

The Transformer architecture is the foundation for almost all modern **Large Language Models (LLMs)** [1]. It is thanks to the power of Deep Learning and Transformer architecture that researchers have been able to build colossal language models, creating a revolution in how computers interact with humans. We will explore LLMs in more detail in the next section.

1.5 Applications Of Llms In The Enterprise.

A Large Language Model (LLM) is a Deep Learning model, typically based on the Transformer architecture, trained on a massive corpus of text data. The most basic objective of its training is to learn how to predict the next word in a sequence of text. For example, given the input "It's a beautiful day, I want to go...", the model learns to predict likely next words such as "for a walk," "fishing," "to the beach," etc.

Although this task may sound simple, when performed on an immense scale, it gives the model extraordinary characteristics. A model is considered "Large" when it possesses the following attributes:

Massive Number of Parameters: Modern LLMs have billions to trillions of parameters. These parameters can be seen as the model's brain, where all the knowledge, syntactic patterns, and semantic understanding it has learned from the data are stored.

Immense Training Data: They are trained on vast datasets of text, including a significant portion of the internet's content such as books, news articles, Wikipedia, and source code. This provides them with a profound knowledge base across almost every domain.

Emergent Abilities: This is the most remarkable and crucial characteristic of LLMs. Due to their immense scale, they spontaneously develop new capabilities for which they were not directly trained. These emergent abilities include text summarization, translation, question-answering, creative writing, programming, and even logical reasoning. It is these emergent abilities that have transformed LLMs from simple word prediction models into versatile artificial brains.

1.6 Concepts of AI Chatbot and AI Agent

1.6.1 Definition of AI Chatbot

Definition: An AI Chatbot is a computer program designed to simulate human conversation through text or voice. The main goal of a chatbot is to "converse" and "provide information" based on a specific knowledge source. It acts as a communication interface, helping users interact with a computer system in a natural way.

1.6.2 Classification of AI Chatbots: Rule-based, Retrieval-based, and Generative AI Chatbots.

Rule-based Chatbots:

How they work: They operate based on a pre-programmed script and logical flow (if-then). They can only answer questions that fall within that script.

Advantages: Easy and inexpensive to develop; answers are consistent and predictable.

Disadvantages: Very rigid; cannot handle out-of-scope questions; lacks context understanding; provides a less natural user experience.

Retrieval-based Chatbots:

How they work: They use information retrieval techniques (like keyword or semantic search) to find the most suitable answer from a database of pre-existing Q&A pairs or documents. The RAG system in this project is a highly advanced form of this type of chatbot.

Advantages: Answers are based on real, reliable data sources and do not fabricate information.

Disadvantages: Cannot generate completely new answers; quality depends on the availability of source data.

Generative Chatbots:

How they work: They use Large Language Models (LLMs) to "generate" new answers based on their vast knowledge learned. ChatGPT is the most typical example.

Advantages: Extremely flexible; can converse on any topic; generates natural and human-like responses.

Disadvantages: Poses a risk of "hallucinations" fabricating non-existent information.

Project Relevance: The RAG Chatbot system intelligently combines the Retrieval-based approach (using RAG for accuracy) and the Generative approach (using an LLM to create natural, coherent answers).

1.6.3 Definition of AI Agent:

AI Agents are artificial intelligence models and algorithms capable of interacting with an environment and making autonomous decisions in the real world without any specific task instructions or guidance from humans. Simply put, AI Agents are personal assistants designed to mimic human intelligence, using AI technologies like **Machine Learning** and **Natural Language Processing** to process and analyze data and perform actions based on user requests and desires. Currently, AI Agents can handle multiple types of inputs simultaneously—such as text, voice, video, audio, and code—thanks to the strong development of **Multimodal AI** and **Generative AI**.

This represents the new future of Artificial Intelligence and is also the next development goal for this project. When AI Agents are widely implemented, infrastructure costs, public models, and the internal data of enterprise departments must have the correct structure and quality. Therefore, the current scope of the RAG Chatbot project for departmental support does not yet apply AI Agent technology.

1.7 Why Rag, Not Fine-Tuning A Pre-Trained Model?

When aiming for a **Large Language Model (LLM)** to "understand" an enterprise's internal data, there are two primary approaches: **Fine-tuning** and **Retrieval-Augmented Generation (RAG)**. Within the scope of this project, our team chose RAG as the core architecture for the following strategic reasons:

Table 3: Comparison RAG and Fine-tuning a Pre-Trained Model

Criterion	Retrieval-Augmented Generation (RAG)	Fine-tuning	Project's Choice
Knowledge Update	Very easy and fast. Only requires updating documents in the database.	Difficult and costly. Requires retraining the entire model when new data is available.	RAG is suitable for a corporate environment where data changes continuously.
Hallucination Prevention	Highly effective. Answers are constrained by the retrieved information, and sources can be cited for	Higher risk. The model can "misremember" or fabricate information if the fine-tuning data	RAG ensures accuracy and reliability, which are crucial factors in an enterprise.

	verification.	is incomplete.	
Cost & Resources	Lower. The primary costs are for API usage and vector storage; it does not require powerful GPUs for training.	Very high. Requires powerful GPU infrastructure and significant time and cost for each training session.	RAG is a feasible and economical choice for a graduation project and for small-to-medium-sized enterprises.
Transparency & Traceability	Very good. It can pinpoint which specific text segment or document was used to generate the answer.	Nearly impossible. Operates as a "black box," making it hard to understand why the model provided a certain answer.	RAG meets the requirement for transparency when information needs to be cross-checked.
Style Modification	Ineffective. RAG primarily changes the model's "knowledge" rather than its "behavior" or "style."	Effective. Can teach the model to speak in a specific tone, style, or format.	The project prioritizes knowledge accuracy over style modification.

CHAPTER 2: SYSTEM ANALYSIS AND DESIGN

2.1 AI System Diagram for Departmental Support Without RAG, and with RAG:

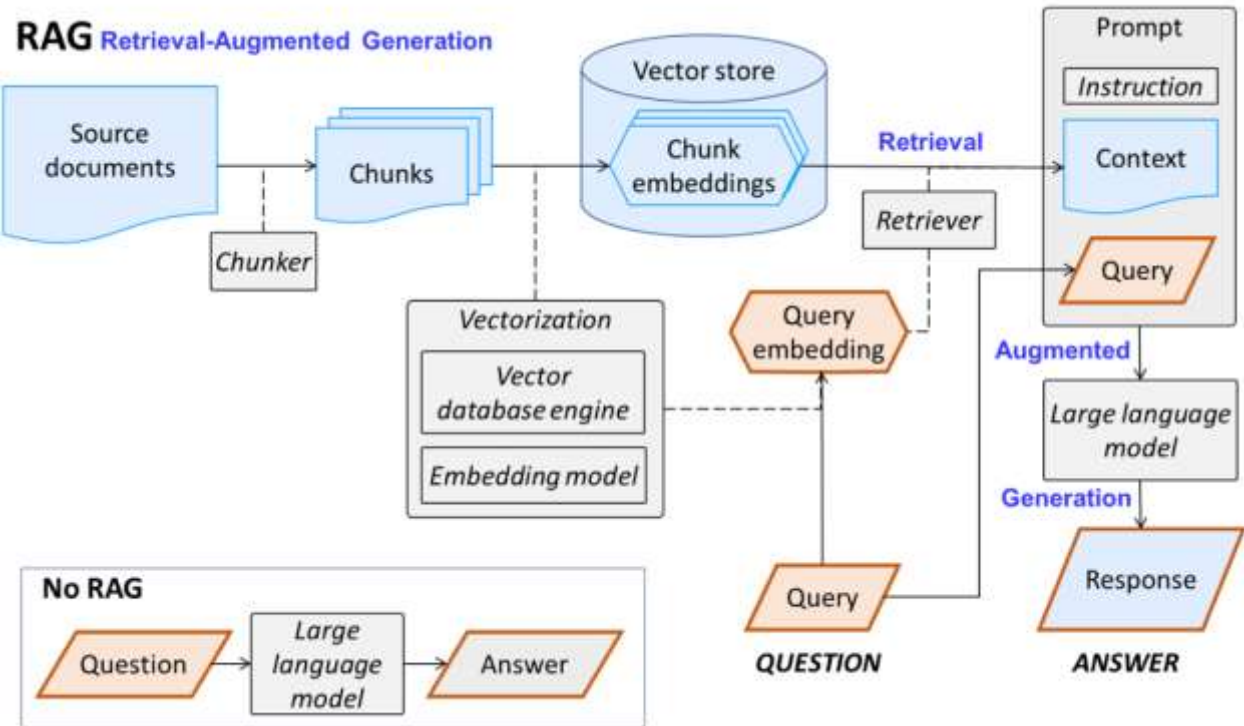


Figure 1: AI System Diagram for Departmental Support Without RAG, and with RAG [3]

2.2 Overall Architecture of RAG and LLM Integration:

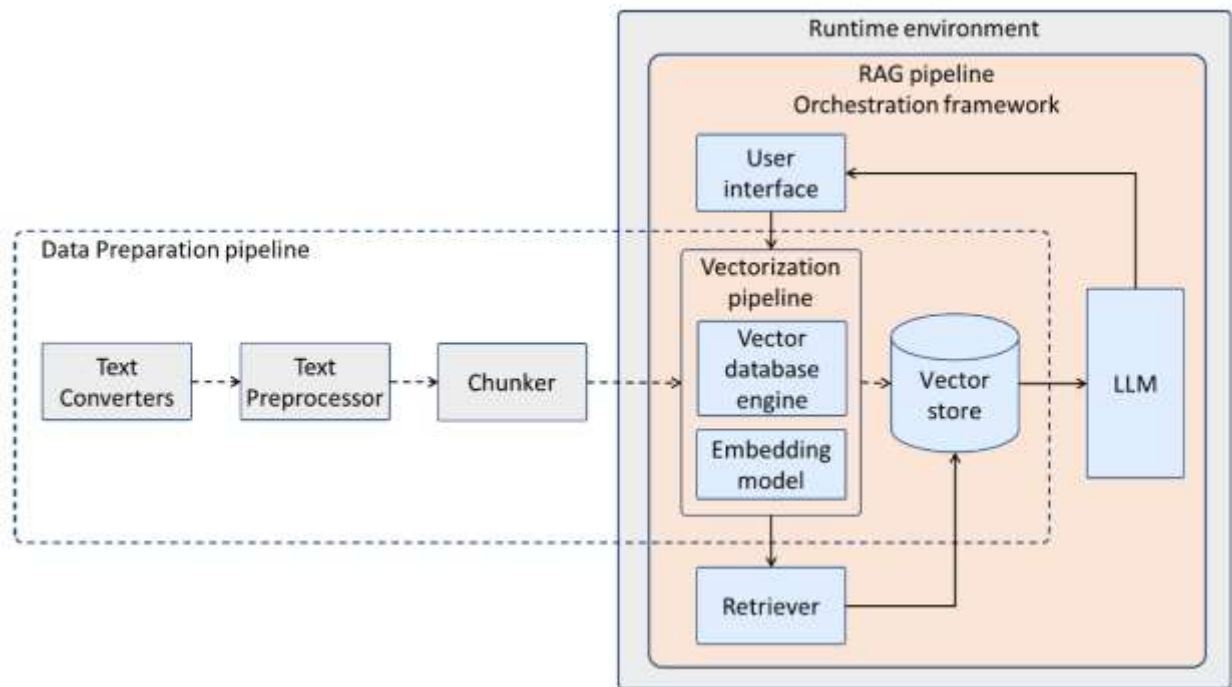


Figure 2: Overall Architecture of RAG and LLM Integration

2.3 Sequence Diagram Corresponding to Use Case

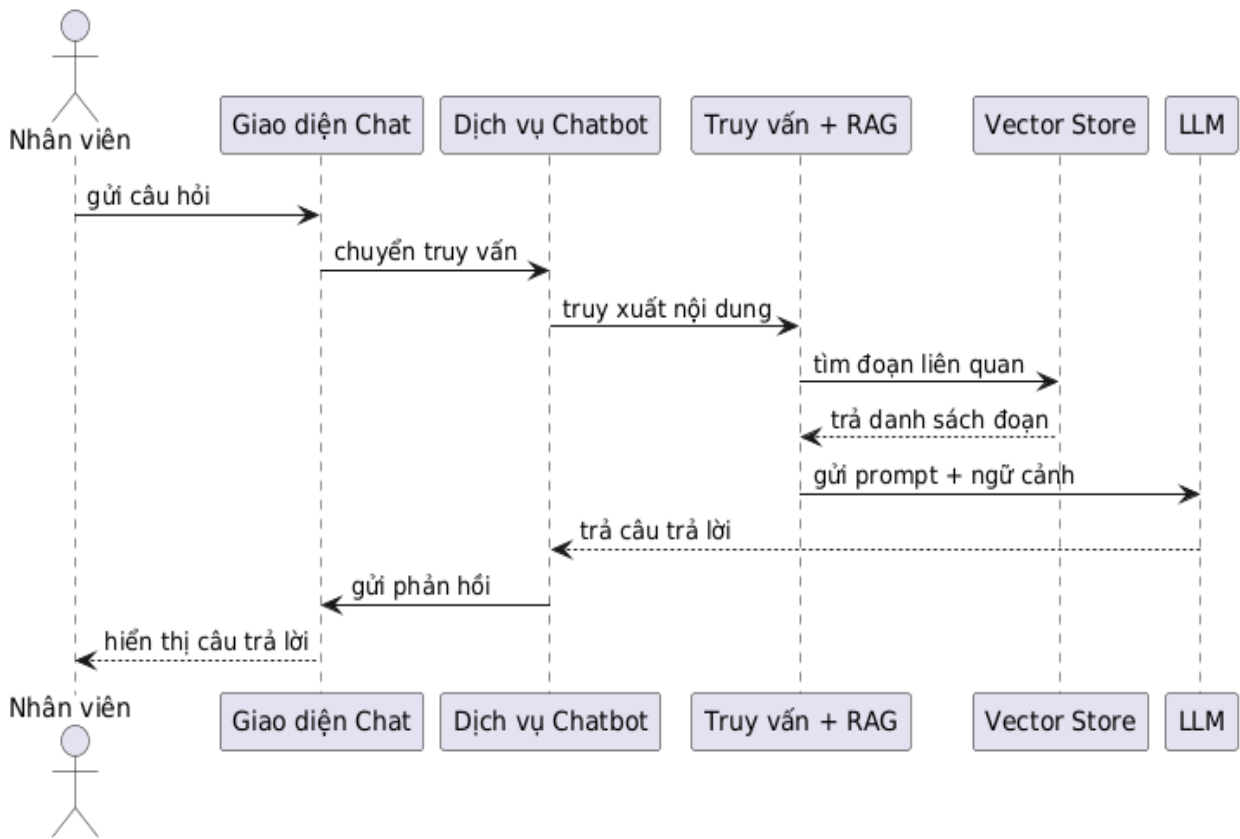


Figure 3: Sequence Diagram

2.4. High-level use-case

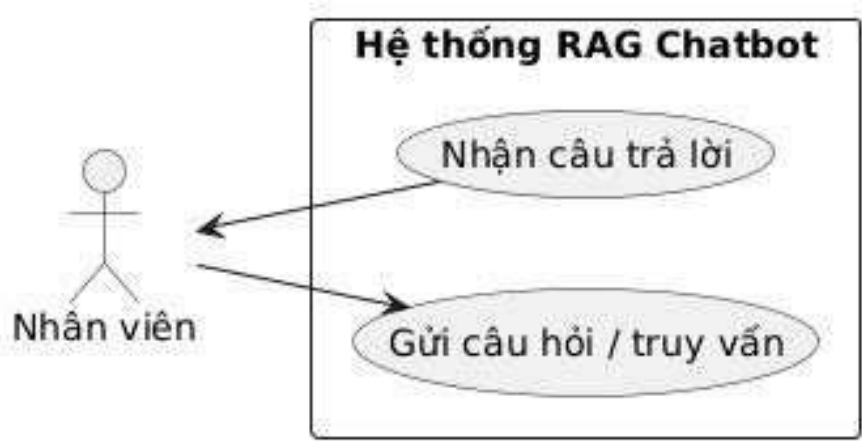


Figure 4: High-level use-case

2.5 Overall Operational Architecture of the RAG Chatbot Project for Departmental Employee Support

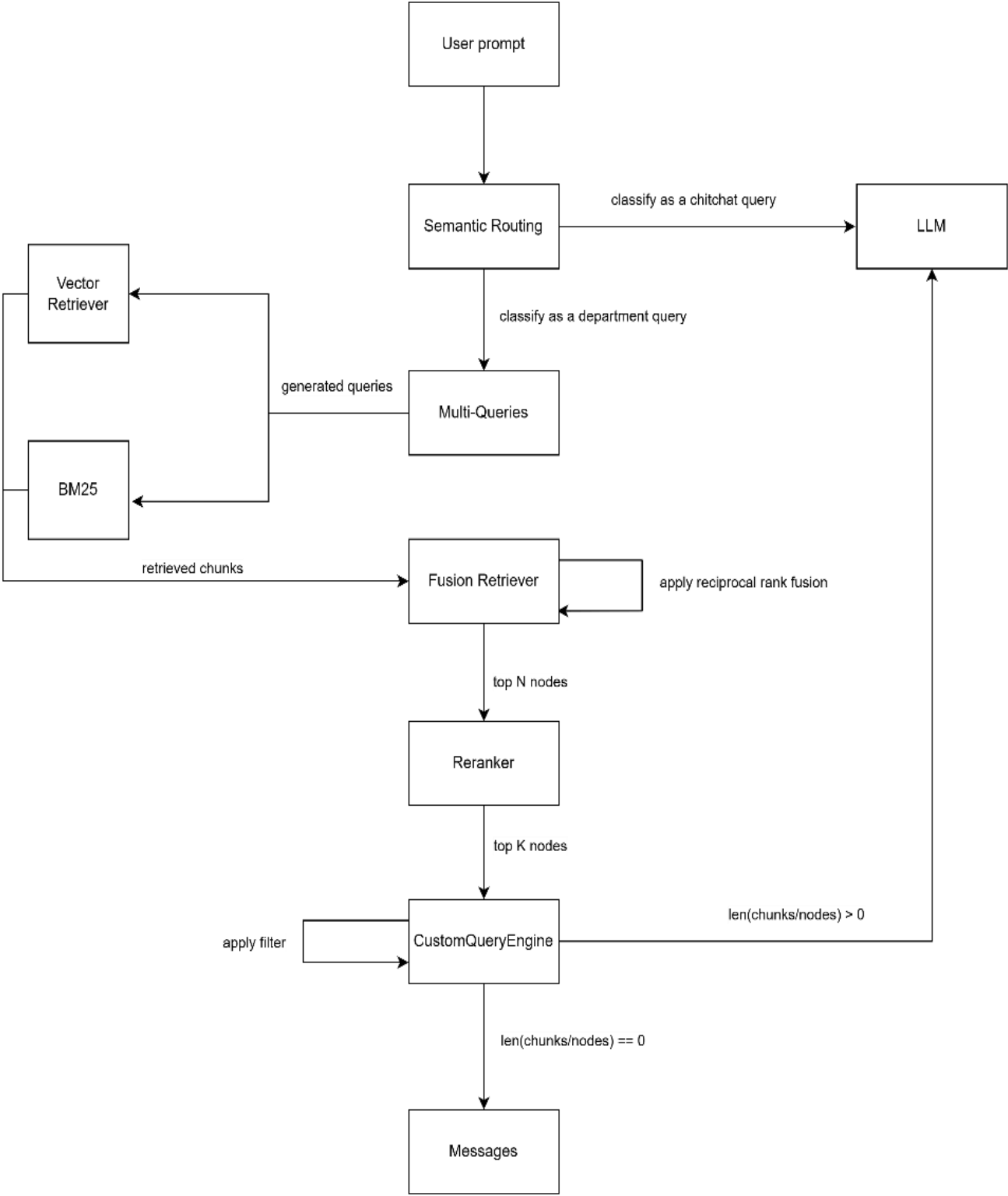


Figure 5: Overall Operational Architecture of the RAG Chatbot Project for Departmental Employee Support [4]

2.5.1 RAG Components and Their Roles

Table 4: RAG Components and Their Roles in Explanation

Component	Role
Runtime environment	Provides the infrastructure and computational resources to host and execute all system components.
Api/end_points	Serves as the main communication gateway for the entire system. It creates an API endpoint (/ask) to receive questions from the user, then forwards the request to the Router for processing and returns the final result to the interface.
BM25	The BM25Retriever applies the BM25 ranking algorithm for lexical search, retrieving the most relevant documents based on keyword frequency and importance. Unlike embeddings, it relies on exact keyword matching. Here, it is customized for Vietnamese by using a specific tokenizer and disabling stemming to improve accuracy.
Config/ embed	Responsible for "translating" text into computer language. It loads a specialized Vietnamese embedding model (AITeamVN/Vietnamese_Embedding), which converts all text into numerical vectors (embeddings), allowing the computer to understand and compare meanings.
Config/llm	Responsible for initializing and configuring the system's LLMs. It securely reads the API key from the environment file and then establishes a connection to the Large Language Model (qwen/qwen3-32b) via the Groq service to generate the final answer.
Config/vector store	Acts as the system's knowledge repository. It uses ChromaDB to create and manage the vector database, where all vectorized documents are persistently stored on your machine.
CustomQueryEngine	custom_query_engine s designed to implement a specialized query engine that controls how users access and receive answers from the RAG system. Its main purpose is to retrieve, rerank, and filter documents by department, ensuring that employees only see information they are authorized to access. It then generates a prompt for the LLM to produce an answer strictly based on the retrieved content. In short, this module customizes the default query flow of LlamaIndex to enforce access control, accuracy, and relevance in enterprise environments.
Chunker	Splits preprocessed text into smaller, manageable segments (chunks) to optimize embedding and retrieval.
Data preparation pipeline	Coordinates the transformation of raw data into structured, chunked, and vectorized format suitable for retrieval and generation.
Fusion Retriever	The Fusion Retriever combines results from different retrievers (e.g., BM25 and vector search). Using techniques like reciprocal rank fusion, it merges and re-ranks the results to provide a more accurate and comprehensive set of documents.
Index builder	Acts as the knowledge base organizer. It performs the entire data processing pipeline: reading documents from files, automatically attaching metadata (like department ID), chunking the text, and then vectorizing each chunk into the Vector Store, making it ready for retrieval. [2]
Multi-Queries	Multi-Queries generates multiple variations of the original query with similar meaning. This expands the search coverage and ensures that more relevant documents can be retrieved, even if the wording differs.

Sematic Routing	Semantic Routing classifies the user's query into different categories (e.g., department-related or general chitchat). It ensures the query is directed to the correct processing flow, improving relevance and efficiency.
Reranker	This is a specialized component that improves the accuracy of search results. It acts as a more sophisticated second-stage filter: after the Retriever fetches a list of potential documents, the Re-ranker re-evaluates and scores the relevance of each document against the original query, then re-sorts them to select only the most relevant results
Retriever/ custom_query_engine	Acts as the coordinator of the Q&A process. It receives the question, directs the Retriever to search for documents, checks the user's departmental access permissions, and then sends the retrieved information to the LLM to generate the final answer.
Retriever/ custom_retriever	The retriever acts as an information retrieval component within the knowledge base. It searches the Vector Store to find the most relevant documents for a given query, while also implementing post-query filtering logic to check metadata and departmental access permissions.
Router/handler	Acts as the information searcher within the knowledge base. It searches the Vector Store to find the most relevant documents to the question, while also implementing post-query filtering logic to check metadata and enforce departmental access permissions.
Router/parser	Defines separate processing flows for each type of request: one flow for general conversation (chitchat) and another to initialize and execute the RAG pipeline for business-related queries that require document lookup.
Router/prompt	Provides the prompt templates to control the LLM's behavior. It defines the processing options (business query, chitchat) and instructs the LLM to classify the user's question, while also requesting the LLM to return its decision in a structured JSON format.
Router/router	Acts as the central router of the system. This module uses the LLM to classify the user's question into different processing flows (e.g., business or chitchat), then routes the request to the appropriate handler function for execution.
Router/schemas	Defines the data schema for the LLM's classification output. Using Pydantic, it creates a "template" (Answer) to ensure that the output from the LLM is always in a consistent and valid format before it is processed.
Retriever	Searches the vector database for the most relevant chunks in response to a query embedding.
Vector database engine	Stores, indexes, and enables efficient similarity search of vector embeddings.
User interface	Enable users to submit queries and view generated responses, facilitating interaction with the system.
utils	A collection of small, reusable helper functions. They perform auxiliary tasks such as string formatting, data extraction, and cleaning to support other core components, which helps keep the codebase organized and tidy.
evluation	Responsible for measuring and validating the quality of the Retriever component. This module automatically generates a sample question answer dataset from the source documents, then calculates standard metrics like Hit Rate and MRR to evaluate its information retrieval performance.

CHAPTER 3: DEPLOYMENT OF THE AI CHATBOT APPLICATION

Chương này nhóm sẽ trình bày chi tiết quá trình xây d

3.1 System Requirements:

To deploy and operate the AI Chatbot application, a standardized system environment is required, including both hardware and software specifications. These requirements ensure sufficient processing performance, model storage capacity, and compatibility among the technological components, thereby guaranteeing the application's stable and efficient operation.

- **Hardware Requirements:** A server with a minimum of a 4-core CPU and 16GB to 32GB of RAM is needed to handle concurrent requests and load the model. A GPU is not mandatory as the system uses the LLM via an API.
- **Software Requirements:** The system operates in a Python 3.9+ environment, utilizing core libraries such as FastAPI, LlamaIndex, and ChromaDB, as well as other supporting libraries.
- **External Services:** A valid API Key from an LLM service provider (e.g., Groq) is required to perform language generation tasks.

3.2 User Interface Demo

3.2.1 Login Interface:

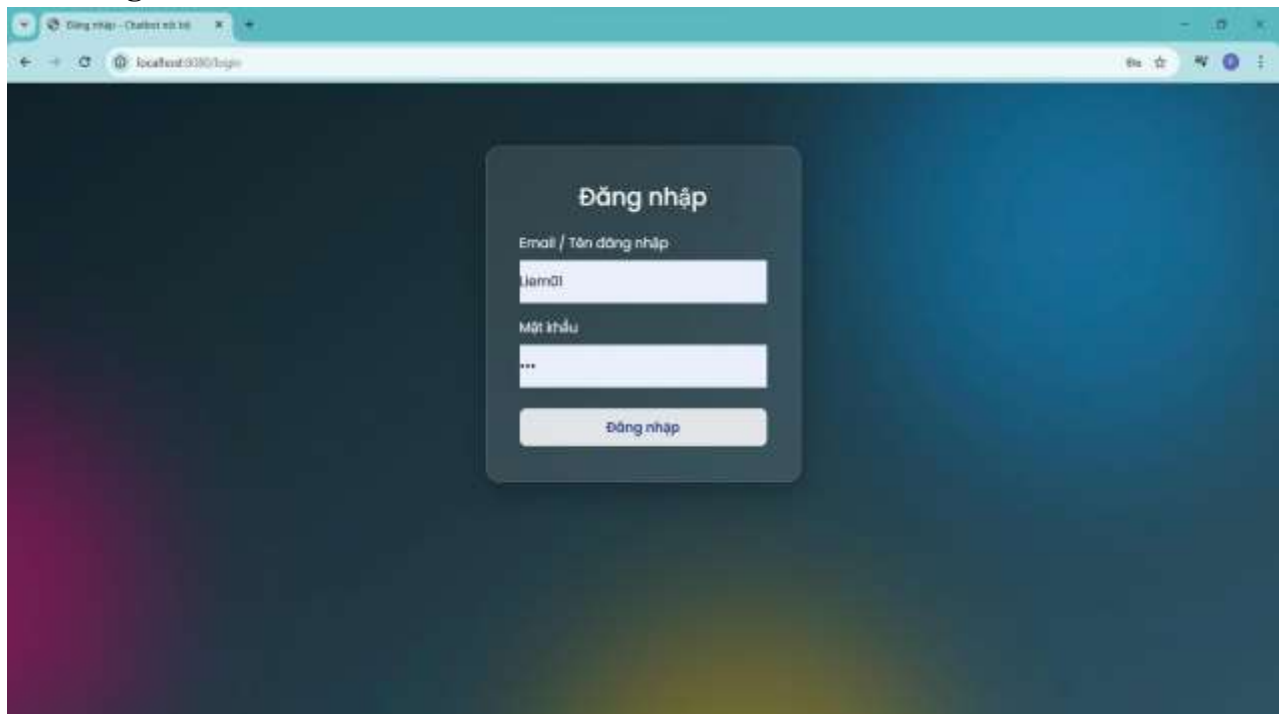


Figure 6: UI Login Interface

3.2.2 Initial Chat Application Interface:

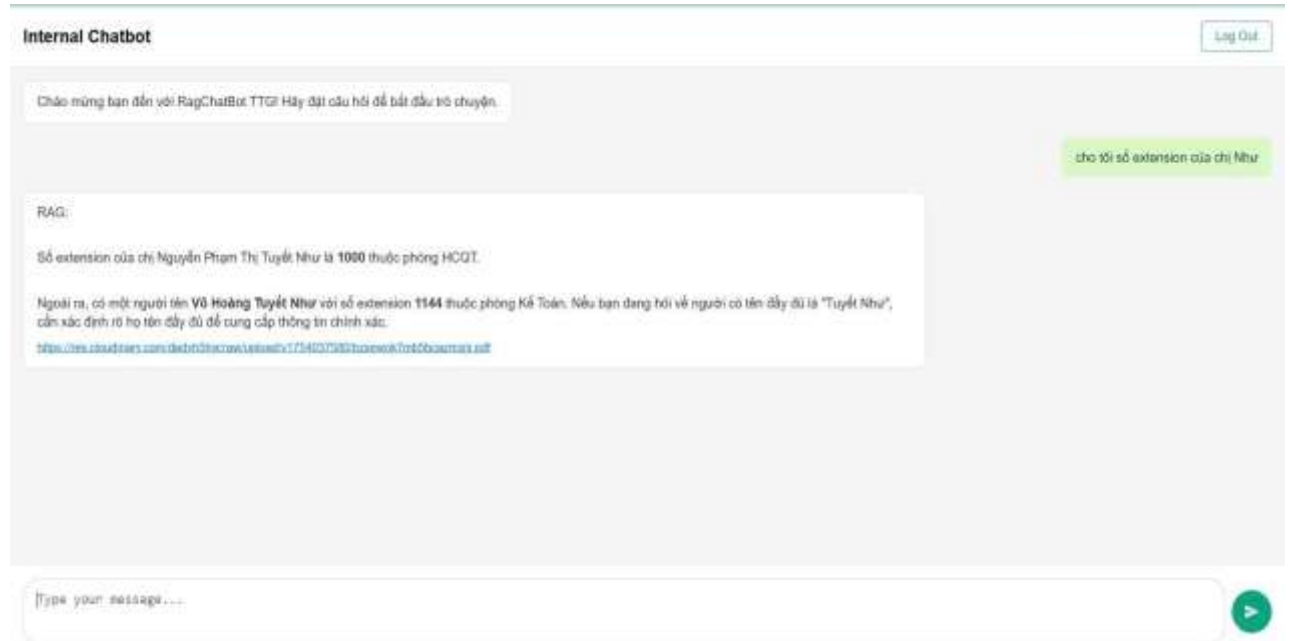


Figure 7: UI Initial Chat Application

3.2.3 Chat Interface with a Query:

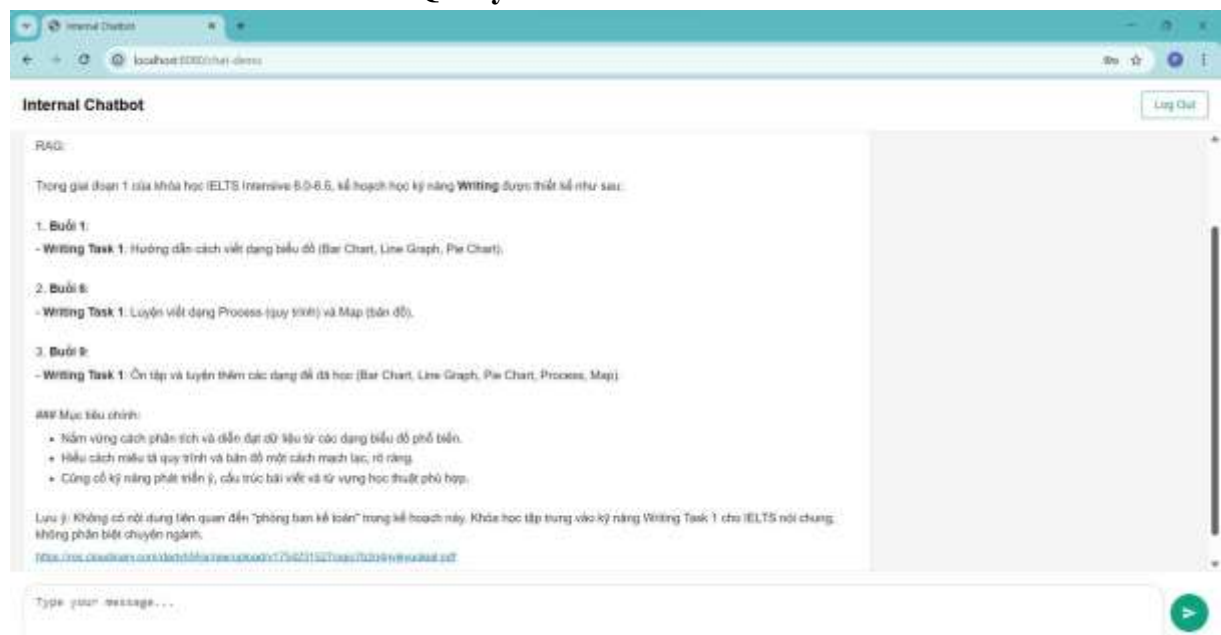


Figure 8: UI Chat Interface with a Query

CHAPTER 4 FUTURE DEVELOPMENT DIRECTIONS

The application of computer science achievements, especially AI and Large Language Models, to create practical and highly applicable tools for enterprises, particularly traditional ones in Vietnam, remains a major digital transformation challenge and a promising, fertile ground. This project concludes at the level of a Chatbot program that leverages RAG and the language and reasoning power of LLMs to build a virtual assistant. This assistant answers questions for employees in various departments, or for new hires undergoing onboarding, regarding technological devices and application usage, drawing from documents provided by the IT Helpdesk department of the team's internship company. From this, we can see that this project has almost limitless potential for development, including:

- **Expanding Role-based Access and Logic for the Web App:** Additional logic and permissions can be implemented for the web application to assign login access for employees in different departments. For example, an employee in the Legal department would be granted access to the specific chatbot workspace for the Legal department, allowing for a more efficient, productive, secure, and personalized experience for each team.
- **Integrating with New AI Models and Agents:** With the continuous development and improvement of new LLMs and AI Agents, this project can be evolved into a web app with broader AI functionalities. This could include features like image generation, video creation via APIs from OpenAI, or leveraging AI Automation tools like n8n or Zapier to work together via Webhooks.
- **Improving the Chatbot Interface and Adding Features:** The chatbot interface can be improved to approach the capabilities of ChatGPT Pro, for example, by adding personalization features, project management functionalities, and Agent-based features.
- **Enhancing the Dashboard:** The dashboard can be improved for deeper administrative control. An administrator could review how employees use the AI chatbot, create accounts for staff based on their roles, and add, delete, or modify employee permissions within the chatbot workspace, similar to the Team version of ChatGPT.

CONCLUSION

After the research and development process for the **AI RAG Chatbot**, despite a short internship period, a tight deadline for the graduation project, and a high-pressure environment with complex new knowledge, our team successfully completed a basic RAG Chatbot web application. We now understand the process of building a chatbot tool, from developing a web app for user management to calling APIs between different services. We have also gained an understanding of the necessary technologies for this project, along with effective time management and teamwork skills. Most importantly, our team has grasped the structure and architecture of an advanced RAG-based Chatbot, the technique for extracting knowledge from IT Helpdesk documents, and how to reduce **hallucinations**—the tendency of LLMs to generate false information. We have also created a system diagram and understood why RAG is a better choice than fine-tuning a pre-trained model. Although the project is still heavily focused on the technical aspects of RAG, with the frontend interface remaining basic and not visually appealing, the future development potential of this topic is immense and highly promising.

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[1] Vaswani, A., et al. (2017). *Attention is All You Need*. *Advances in Neural Information Processing Systems*, 30

[2] LlamaIndex Documentation:

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[3] Luong, Q. (2025). *Offline open-source RAG-based AI chatbot for Finnish-language software documentation on low-end hardware: Modular model-agnostic framework and AI judge for multilingual use*. *Bachelor's Thesis, Degree Programme in Computer Applications*.

[4] *Retrieval-Augmented Generation for Large Language Models: A Survey*
<https://arxiv.org/abs/2312.10997>